

PERFORMANCE TESTING

Visualization Tool for Electric Vehicle Charge and Range Analysis

1. Introduction

Performance testing was conducted to evaluate the responsiveness, stability, and usability of the Visualization Tool for Electric Vehicle Charge and Range Analysis. The testing focused on verifying data loading, dashboard performance, SQL query execution, and user interaction with Tableau visualizations.

2. Test Objectives

- Verify dashboard loading performance.
- Ensure SQL queries execute correctly.
- Validate accuracy of displayed data.
- Check responsiveness of filters and charts.
- Ensure smooth navigation between dashboards and stories.

3. Test Environment

Parameter	Details
Operating System	Windows 10
Database	MySQL Workbench
Visualization Tool	Tableau Public
RAM	8 GB
Processor	Intel Core i3
Browser	Google Chrome

4. Test Results

Test Case	Expected Result	Actual Result	Status
Dashboard Loading	Dashboard opens correctly	Successfully opened	Pass
SQL Data Retrieval	Correct data displayed	Correct data displayed	Pass
Interactive Filters	Filters respond properly	Working correctly	Pass
Dashboard Navigation	All sheets open correctly	Working correctly	Pass
Story Navigation	Story displays correctly	Working correctly	Pass
Visualization Accuracy	Charts show correct data	Verified	Pass

5. Performance Summary

The dashboard performed efficiently during testing. SQL queries retrieved the required data successfully, Tableau dashboards loaded without significant delay, and interactive filters responded correctly. The overall performance met the project requirements.

6. Conclusion

The Visualization Tool for Electric Vehicle Charge and Range Analysis successfully passed all performance tests. The application demonstrated reliable data retrieval, accurate visualization, and smooth user interaction, making it suitable for electric vehicle data analysis.